Discussion 9

1. Simple Constrained Optimization Problem with Duality

Consider the optimization problem

$$\min_{x_1,x_2\in\mathbb{R}} \quad f(x_1,x_2) \tag{1}$$

$$\text{s.t.} \quad 2x_1 + x_2 \geq 1 \tag{2}$$

$$x_1 + 3x_2 \geq 1 \tag{3}$$

$$x_1 \geq 0, \tag{4}$$

$$x_2 \geq 0 \tag{5}$$

(a) Express the Lagrangian of the problem $\mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4)$.

Solution:

$$\mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = f(x_1, x_2) + \lambda_1(-2x_1 - x_2 + 1) + \lambda_2(1 - x_1 - 3x_2) - \lambda_3 x_1 - \lambda_4 x_2 \tag{6}$$

(b) Show that $\mathcal{L}$ is concave in $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$.

Solution: $-\mathcal{L}$ is convex in $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ as a affine function of $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$. So $\mathcal{L}$ is concave in $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$

(c) Express the dual function of the problem, and show that it is concave.

Solution: $g(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = \min_{x_1,x_2} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4)$.

We can show that by showing that $-g$ is convex.

$$-g(\vec{\lambda}) = -\min_{x_1,x_2} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) \tag{7}$$

$$= \max_{x_1,x_2} -\mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) \tag{8}$$

When (x_1, x_2) is fixed, the function $-\mathcal{L}$ is linear in $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$, therefore convex.

Because the max of convex functions is convex, $-g$ is convex. Therefore g is concave.

(d) Assume f is convex. Show that $\mathcal{L}$ is convex in (x_1, x_2).

Solution: $\mathcal{L}$ is convex in (x_1, x_2) because it is the sum of convex functions.

(e) Denoting $\mathcal{X} = \{(x_1, x_2) \mid 2x_1 + x_2 \geq 1,\ x_1 + 3x_2 \geq 1,\ x_1 \geq 0,\ x_2 \geq 0\}$, show that

$$\max_{\lambda_1\geq 0,\lambda_2\geq 0,\lambda_3\geq 0,\lambda_4\geq 0} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = \begin{cases} f(x_1, x_2) & \text{if } (x_1, x_2) \in \mathcal{X} \\ +\infty & \text{otherwise} \end{cases} \tag{9}$$

Solution: Let's just do it for λ_4:

$$\max_{\lambda_4 \geq 0} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = \max_{\lambda_4 \geq 0} \left(f(x_1, x_2) + \lambda_1(-2x_1 - x_2 + 1) + \lambda_2(1 - x_1 - 3x_2) - \lambda_3 x_1 - \lambda_4 x_2\right) \tag{10}$$

$$= f(x_1, x_2) + \lambda_1(-2x_1 - x_2 + 1) + \lambda_2(1 - x_1 - 3x_2) - \lambda_3 x_1 + \max_{\lambda_4 \geq 0} -\lambda_4 x_2 \tag{11}$$

$$\max_{\lambda_4 \geq 0} -\lambda_4 x_2 = \begin{cases} 0 & \text{if } x_2 \geq 0 \\ +\infty & \text{otherwise} \end{cases} \tag{12}$$

One can show the same results for λ_1, λ_2 and λ_3, resulting in:

$$\max_{\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = \begin{cases} f(x_1, x_2) & \text{if } (x_1, x_2) \in \mathcal{X} \\ +\infty & \text{otherwise} \end{cases} \tag{13}$$

(f) Conclude that $\min_{x_1, x_2 \in \mathbb{R}} \max_{\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = \min_{(x_1, x_2) \in \mathcal{X}} f(x_1, x_2)$.

Solution:

$$\min_{x_1, x_2} \max_{\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = \min_{x_1, x_2} \begin{cases} f(x_1, x_2) & \text{if } (x_1, x_2) \in \mathcal{X} \\ +\infty & \text{otherwise} \end{cases} \tag{14}$$

$$= \min_{(x_1, x_2) \in \mathcal{X}} f(x_1, x_2) \tag{15}$$

(g) Assuming f is convex, formulate the first order condition on $\mathcal{L}$ (i.e., $\nabla_{x_1, x_2} \mathcal{L}(x_1^\star, x_2^\star, \lambda_1, \lambda_2, \lambda_3, \lambda_5) = 0$) as a function of ∇f and $\lambda_1, \lambda_2, \lambda_3$ and λ_4 to solve:

$$\min_{x_1, x_2} \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4) \tag{16}$$

Solution:

$$\nabla_{x_1, x_2} \mathcal{L}(x_1^\star, x_2^\star, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = 0 \tag{17}$$

$$= \nabla_{x_1, x_2} f(x_1^\star, x_2^\star) + \begin{bmatrix} -2\lambda_1 - \lambda_2 - \lambda_3 \\ -\lambda_1 - 3\lambda_2 - \lambda_4 \end{bmatrix} \tag{18}$$

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2. Dual Space

Let $\|\cdot\|$ be a norm on $\mathcal{X} = \mathbb{R}^n$. Define the *dual space* of $\mathcal{X}$ to be the set of linear functions defined on $\mathcal{X}$, and denote it by

$$\mathcal{X}^* := \{f : \mathcal{X} \to \mathbb{R} \mid f \text{ is a linear function}\}.$$

Show that $\mathcal{X}^*$ is a vector space (with scalar field $\mathbb{R}$). In particular, verify that for any $f_1, f_2 \in \mathcal{X}^*$ and scalars $\alpha, \beta \in \mathbb{R}$, the function $\alpha f_1 + \beta f_2$ remains in $\mathcal{X}^*$.

Then, show that

$$\|f\|_{op} := \max_{x:\|x\|\leq 1} f(x), \quad f \in \mathcal{X}^*$$

defines a norm on $\mathcal{X}^*$.

Solution:

Closure under linear combinations

If f_1, f_2 are linear functions and $\alpha, \beta \in \mathbb{R}$, then for any $x, y \in \mathcal{X}$ and scalars $a, b \in \mathbb{R}$:

$$\begin{aligned}
\bar{f}(ax + by) &= \alpha f_1(ax + by) + \beta f_2(ax + by) \\
&= \alpha(af_1(x) + bf_1(y)) + \beta(af_2(x) + bf_2(y)) \\
&= a(\alpha f_1(x) + \beta f_2(x)) + b(\alpha f_1(y) + \beta f_2(y)) \\
&= a\bar{f}(x) + b\bar{f}(y).
\end{aligned}$$

Since $\bar{f}$ preserves vector addition and scalar multiplication, it is a linear function, meaning $\bar{f} \in \mathcal{X}^*$. Therefore, $\mathcal{X}^*$ is a vector space.

Proof that the operator norm is a norm in the vector space of linear functions

Now, we show that $\|f\|_{op}$ is a norm on $\mathcal{X}^*$. Since $\mathcal{X} = \mathbb{R}^n$, any linear function $f \in \mathcal{X}^*$ can be represented as an inner product. Let e_i be the i-th standard basis vector and define $f_i := f(e_i)$. By the linearity of f, for any $x \in \mathbb{R}^n$, we have:

$$f(x) = f\left(\sum_{i=1}^n x_i e_i\right) = \sum_{i=1}^n x_i f(e_i) = \mathbf{f}^T x,$$

where $\mathbf{f} = [f_1, \ldots, f_n]^T \in \mathbb{R}^n$.

Substituting this representation into the definition of the operator norm yields:

$$\|f\|_{op} = \max_{x:\|x\|\leq 1} \mathbf{f}^T x.$$

We recognize this exactly as the definition of the dual norm $\|\mathbf{f}\|_*$ on $\mathbb{R}^n$, which is known to satisfy all the properties of a valid norm. Because the mapping $f \mapsto \mathbf{f}$ is a linear isomorphism in f, and the properties of a norm only reference aspects of the linear structure of the set over which it is defined, we can conclude that $\|f\|_{op}$ inherits all norm properties from $\|\mathbf{f}\|_*$. Thus, the fact that the dual norm is a norm is sufficient to show that the operator norm $\|\cdot\|_{op}$ for linear functions $f : \mathcal{X} = \mathbb{R}^n$ is a norm, concluding the proof.

Suppose, however, that we do not know that the dual norm is a norm. Then, we must check that the three properties of a norm are satisfied for either the dual norm or $\|\cdot\|_{op}$, and the isomorphism that we have discussed will imply that both norms are indeed norms. This includes the triangle inequality, absolute homogeneity, and positive definiteness. The proofs of these will all have the same structure for both norms because they only reference properties that are in direct correspondence via our isomorphism.

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Triangle inequality

We must prove that for all linear functions f, $\|f+g\|_{op} \leq \|f\|_{op} + \|g\|_{op}$.

For any x in the feasible set,

$$\begin{aligned}
&f(x) + g(x) \\
&\qquad \leq f(x) + \max_{y:\|y\|\leq 1} g(y) && \text{by def. of max, because } \|x\| \leq 1 \\
&\qquad \leq \max_{z:\|z\|\leq 1} f(z) + \max_{y:\|y\|\leq 1} g(y) && \text{by def. of max, because } \|x\| \leq 1 \\
&\qquad = \|f\|_{op} + \|g\|_{op} && \text{by def. of } \|\cdot\|_{op}
\end{aligned}$$

Since this inequality holds for all x in the feasible set, it holds for the maximum over all such x – that is, $\|f+g\|_{op} \leq \|f\|_{op} + \|g\|_{op}$.

Absolute homogeneity

Then, we should prove absolute homogeneity: for all linear functions f and real numbers α, $\|\alpha f\|_{op} = |\alpha|\|f\|_{op}$. In the following, let $\text{sgn}(\alpha)$ denote 1 if $\alpha \geq 0$ and -1 if $\alpha < 0$.

$$\begin{aligned}
\|\alpha f\|_{op} &= \max_{x:\|x\|\leq 1} \alpha f(x) && \text{by def. of } \|\cdot\|_{op} \\
&= \max_{x:\|x\|\leq 1} |\alpha|\text{sgn}(\alpha) f(x) && \text{by def. of sgn} \\
&= |\alpha| \max_{x:\|x\|\leq 1} f(\text{sgn}(\alpha)x) && \text{by linearity of } f \\
&= |\alpha| \max_{x:\|x\|\leq 1} f(x) && \text{because multiplication by } \text{sgn}(\alpha) \text{ is a bijection of the feasible set with itself} \\
&= |\alpha|\|f\|_{op} && \text{by def. of } \|\cdot\|_{op}
\end{aligned}$$

Positive definiteness

We must show that $\|f\|_{op} \geq 0$ and that $\|f\|_{op} = 0 \iff f = 0$, that is, the norm of f is zero if and only if f is the zero function. Positive definiteness therefore gives us three things to prove: Nonnegativity, and both directions of this if-and-only-if.

Since $\vec{0}$ is in the feasible set, and $f(\vec{0}) = 0$ by linearity of f, the maximum of f over the feasible set is at least zero. This proves nonnegativity.

The leftward implication $f = 0 \implies \|f\|_{op} = 0$ is immediate from the definitions: $\|f\|_{op} = \max_{x:\|x\|\leq 1} f(x) = \max_{x:\|x\|\leq 1} 0 = 0$.

For the rightward implication $\|f\|_{op} = 0 \implies f = 0$, we can prove the contrapositive. Suppose $f \neq 0$. Then there is some y such that $f(y) \neq 0$, and $y' := \frac{y}{\|y\|}$ is in the feasible set, and by linearity $f(y') \neq 0$. If $f(y') > 0$ then we are done, because $\|f\|_{op} = \max_{x:\|x\|\leq 1} f(x) \geq f(y') > 0$. Otherwise, we can use the fact that the feasible set is closed under negation to find that $\|f\|_{op} = \max_{x:\|x\|\leq 1} f(x) \geq f(-y') > 0$.

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